

公式的矩阵推导.

公式 (3):

原式: $\begin{cases} \dot{x}_i(t) = v_i(t) \\ \dot{v}_i(t) = v_i(t) - \sum_{j=1}^n a_{ij} [(x_i(t) - x_j(t) - r_{ij}) + \beta (v_i(t) - v_j(t))] \\ \quad - k_i [x_i(t) - x_c(t) - r_i] + \gamma (v_i(t) - v_c(t)) \end{cases}$

• 所用矩阵

邻接矩阵 $A = A = [a_{ij}] \in \mathbb{R}^{n \times n}$

度矩阵 D : 对角线元素为 $d_i = \sum_{j=1}^n a_{ij}$ 的对称矩阵.

拉普拉斯矩阵 L : $L = D - A \in \mathbb{R}^{n \times n}$

邻居连接矩阵 $K = \text{diag}(k_1, \dots, k_n) \in \mathbb{R}^{n \times n}$

• 推导.

1. 定义位置矩阵 $X = [x_1^T, x_2^T, x_3^T, \dots, x_n^T]$
全局速度矩阵 $V = [v_1^T, v_2^T, v_3^T, \dots, v_n^T]$
期望间距矩阵 $R = [r_1^T, \dots, r_n^T]$
 $\rightarrow \begin{cases} \in \mathbb{R}^{n \times 2} \end{cases}$

定义全局误差向量: $\tilde{X} = [\tilde{x}_1, \dots, \tilde{x}_n]^T = X - \frac{1}{n} X \mathbf{1}^T - R$
 $\tilde{V} = [\tilde{v}_1, \dots, \tilde{v}_n]^T = V - v_c^T$

2.
$$\begin{aligned} \sum_{j=1}^n a_{ij} [x_i(t) - x_j(t)] &= \left(\sum_{j=1}^n a_{ij} \right) \cdot x_i(t) - \sum_{j=1}^n (a_{ij} x_j(t)) \\ &= D \cdot X - A X \\ &= L X \end{aligned}$$

同理 $r_{ij} = r_i - r_j$

$$\therefore \sum_{j=1}^n a_{ij} [x_i(t) - x_j(t) - r_{ij}] = L (X - R)$$

$$\textcircled{2} \quad L(X-R) = L\tilde{X}$$

$$LV = L\tilde{V}$$

证:

$$\sum a_{ij} (v_i - v_j)$$

$$= \sum a_{ij} ((v_i + v_k) - (v_j + v_k))$$

$$\therefore v_i, v_j \Rightarrow v_k$$

$$\therefore LV_k = 0$$

$$\text{同理 } LK_k = 0$$

$$\therefore \text{总体原式} = \dot{\tilde{V}} = -(L+K)\tilde{X} - (\beta L + \gamma K)\tilde{V}$$